Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/27/2004
PI Name:	Kostelecky, V. Alan Ph.D.		
Project Title:	Theoretical Studies of Lorentz and CPT Symmetry		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kostelec@indiana.edu	**Fax:**	FY 812-855-5533
PI Organization Type:	UNIVERSITY	**Phone:**	812-855-1485
Organization Name:	Indiana University		
PI Address 1:	Physics Department, Indiana University		
PI Address 2:			
PI Web Page:	http://www.physics.indiana.edu/		
City:	Bloomington	**State:**	IN
Zip Code:	47405	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2003	**End Date:**	04/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2914		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth"s gravity		
Task Description:	A general extension of the standard model of particle physics incorporating gravity allows for the possibility of Lorentz and CPT violation while otherwise preserving conventional physics. Sensitive experiments searching for predicted effects include ones that may be advantageously performed in a microgravity environment. To develop and study theoretically possible violations of spacetime symmetry and to identify sensitive experimental tests for the associated effects. High-sensitivity tests of fundamental symmetries can shed light on physics at the Planck scale and could offer experimental support for an underlying theory such as strings.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	A detailed study was performed of space-based experiments offering sensitivity to Lorentz and CPT violation. This work is published in Physical Review D. It provides a classification of clock sensitivities and presents explicit formulae for time variations arising in space-based experiments. An analysis was presented of data from a clock-comparison experiment performed by R. Walsworth's group at the Harvard-Smithsonian Center for Astrophysics. This represents results of a search for sidereal annual variations in the frequency difference between xenon-129 and helium-3 masers. A comprehensive study of the neutrino sector of the SME was undertaken, described in published work. Possible signals in existing and future experiments are studied. A detailed investigation of the role of the gravitational sector in the SME was performed. This is published in Physical Review D. The incorporation of Lorentz and CPT violation into Einstein's General Relativity and other theories of gravity based on Riemann-Cartan geometries is established. Various applications to ground- and space-based experiments are suggested.
Bibliography Type:	Description: (Last Updated: 03/05/2008)
Articles in Peer-reviewed Journals	R.Bluhm,V.A.Kostelecky,C.Lane,N.Russell "Probing Lorentz and CPT Symmetry with Space-Based Experiments" Phys Rev D 68, 125008 (2003) , Dec-2003
Articles in Peer-reviewed Journals	F.Cane et al. "Bound on Lorentz- and CPT-Violating Boost Effects for the Neutron" physics/0309070 submitted , Aug-2003
Articles in Peer-reviewed Journals	V.A. Kostelecky and M.Mewes "Lorentz and CPT Violation in Neutrinos" Phys Rev D 69, 016005 (2004) , Jan-2004
Articles in Peer-reviewed Journals	V.A. Kostelecky "Gravity, Lorentz Violation, and the Standard Model" Phys. Rev. D 69, 105009 (2004) , May-2004